AIST3030/ESTR3114: Homework 1

Deadline: October 07, 3:00 pm on Gradescope

Please log in to Gradescope using your CUHK email address to submit your homework solutions.

Problem 1: Answer the following questions on vectors and matrices.

1. Plot the 2-dimensional regions $\{\mathbf{x} = [x_1, x_2] : \|\mathbf{x}\|_p \leq 1\}$ for $p = 1, 2, \infty$.
2. Let S denote the subspace of $\mathbb{R}^5$ spanned by the following two vectors

$$\mathbf{x}_1 := [2, 3, 2, -1, 0]^\top, \quad \mathbf{x}_2 := [1, -2, 1, -2, 1]^\top$$

   (a) Find the inner product $\langle \mathbf{x}_1, \mathbf{x}_2 \rangle$. What is the dimension of subspace S?

   (b) Find the angle between $\mathbf{x}_1, \mathbf{x}_2$. Are $\mathbf{x}_1, \mathbf{x}_2$ orthogonal vectors?

   (c) Is the set $\{\mathbf{x}_1, \mathbf{x}_2\}$ an orthnormal basis for S? If not, find an orthonormal basis for S.

3. Consider the following two matrices

$$A = \begin{bmatrix} 7 & 3 \\ 3 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 5 \\ 0 & 5 & 3 \end{bmatrix}$$

   ,

   (a) Find the rank and determinant of A and B.

   (b) Find the eigenvalues and eigenvectors of A and B.

Problem 2 : As discussed in class, Matrix A is called diagonal if all its off-diagonal entries are zero, i.e. A's (i, j)th entry $a_{i,j} = 0$ for every $i \neq j$. We use the following notation to represent an $n \times n$ diagonal matrix with diagonal entries $a_1, a_2, \ldots, a_n$:

$$\text{diag}\{a_1, a_2, \ldots, a_n\}$$

1. Under what conditions on $a_1, a_2, \ldots, a_n$, the diagonal matrix $\text{diag}\{a_1, a_2, \ldots, a_n\}$ is invertible? What will be the inverse of $\text{diag}\{a_1, a_2, \ldots, a_n\}$?
2. Under what conditions on $a_1, a_2, \ldots, a_n$, the diagonal matrix $\text{diag}\{a_1, a_2, \ldots, a_n\}$ is positive semi-definite? Under that condition, find a matrix A such that $\text{diag}\{a_1, a_2, \ldots, a_n\} = AA^\top$.
3. Find the eigenvectors and eigenvalues of $\text{diag}\{a_1, a_2, \ldots, a_n\}$. Use the eigenvalues to find the determinant of $\text{diag}\{a_1, a_2, \ldots, a_n\}$ and also to verify your solutions to Parts 1,2.

Problem 3: Recall that we use $\text{Rank}(A)$ to denote the rank of matrix A.

1. Consider two column vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. In class, we discussed that the inner product can be written as matrix multiplication
$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top \mathbf{y},$$
implying that $\mathbf{x}^\top \mathbf{y}$ is a scalar number or a $\mathbf{1 \times 1}$ **matrix**. Consider the other transposition order $\mathbf{x}\mathbf{y}^\top$ where we put transpose on the second vector. How many columns and rows does matrix $\mathbf{x}\mathbf{y}^\top$ have? What is the rank of this matrix $\text{Rank}(\mathbf{x}\mathbf{y}^\top)$?

2. Suppose $A = B + C$ is the sum of matrices B and C. Prove that
$$\text{Rank}(A) \leq \text{Rank}(B) + \text{Rank}(C)$$
Hint: Consider a basis for the range of B and C. Can we write every vector in A's range with a linear combination of the two basis sets?

3. Suppose $A = BC$, where $A \in \mathbb{R}^{m \times n}$ is the product of matrices $B \in \mathbb{R}^{m \times k}$ and $C \in \mathbb{R}^{k \times n}$. Prove that $\text{Rank}(A) \leq k$.
Hint: Consider $B = [\mathbf{b}_1 | \mathbf{b}_2 | \cdots | \mathbf{b}_k]$ and $C = [\mathbf{c}_1^\top; \mathbf{c}_2^\top; \cdots; \mathbf{c}_k^\top]$ where $\mathbf{b}_i$ is the ith column of B and $\mathbf{c}_j^\top$ is the jth row of C. With this in mind, can we use parts 1 and 2 to show part 3?

4. Again, assume that $A = BC$, where $A \in \mathbb{R}^{m \times n}$ is the product of matrices $B \in \mathbb{R}^{m \times k}$ and $C \in \mathbb{R}^{k \times n}$. Prove that
$$\text{Rank}(A) \leq \min\{\text{Rank}(B), \text{Rank}(C)\}$$
Does the above inequality imply $\text{Rank}(A) \leq k$?

Problem 4: In this question, you find the gradient and Hessian of some norm-based and quadratic functions. Note that the norm functions in this problem are the Euclidean norm.

1. Let $f(\mathbf{x}) = (\mathbf{x} + \mathbf{b})^\top A(\mathbf{x} + \mathbf{b})$ for a fixed matrix A and vector $\mathbf{b}$. Find $\nabla f(\mathbf{x})$ and $H_f(\mathbf{x})$.
2. Let $f(\mathbf{x}) = \|A\mathbf{x} - \mathbf{b}\|^2$ for a fixed vector $\mathbf{b}$ and matrix A. Find $\nabla f(\mathbf{x})$ and $H_f(\mathbf{x})$.
3. Let $f(\mathbf{x}) = \|A\mathbf{x} - \mathbf{b}\|$ for a fixed vector $\mathbf{b}$ and matrix A. Find $\nabla f(\mathbf{x})$ and $H_f(\mathbf{x})$.